

The question is then how assumption (6.12) can be justified. The wide-spread appearance of  $1/f$  noise can only happen when scale-invariant distribution of relaxation times are ubiquitous, viz if they were self-organized. The  $1/f$  noise therefore constitutes an interesting motivation for the search of possible mechanisms leading to self-organized criticality.

### 6.3 Cellular Automata

Cellular automata are finite state lattice systems with discrete local update rules.

$$z_i \rightarrow f_i(z_i, z_{i+\delta}, \dots), \quad z_i \in [0, 1, \dots, n], \quad (6.14)$$

where  $i + \delta$  denote neighboring sites of site  $i$ . Each site or “cell” of the lattice follows a prescribed rule evolving in discrete time steps. At each step the new value for a cell depends only on the current state of itself and on the state of its neighbors. Cellular automata differ from general dynamical networks in two aspects.<sup>5</sup>

- The update functions are all identical:  $f_i() \equiv f()$ , viz they are translational invariant.
- The number  $n$  of states per cell is usually larger than two, the boolean case.

Cellular automata can give rise to exceedingly complex behavior despite their deceptively simple dynamical structure. We note that cellular automata are always updated synchronously and never sequentially or randomly. The state of all cells is renewed simultaneously.

**Number of Update Rules** The number of possible update rules is exponentially large. For a two-dimensional model, the square lattice, we consider cells taking one of two possible states,

$$z_i = 0 \quad (\text{dead}), \quad z_i = 1 \quad (\text{alive}).$$

For a basic type of rules the evolution of a given cell to the next time step depends only on the current state of the cell and on the values of each of its eight nearest neighbors. In this case there are

$$2^9 = 512 \text{ configurations}, \quad 2^{512} = 1.3 \times 10^{154} \text{ possible rules},$$

since any one of the 512 configurations can be mapped independently to ‘live’ or ‘dead’. For comparison note that the age universe is of the order of  $3 \times 10^{17}$  seconds.

<sup>5</sup> A particularly important class of dynamical networks are studied in Chap. 7.

**Totalistic Update Rules** It clearly does not make sense to explore systematically the consequences of arbitrary updating rules. One simplification is to consider a mean field approximation that results in a subset of rules termed “totalistic”. For mean field rules, the new state of a cell depends only on the total number of living neighbors, besides its own state. An eight-cell neighborhood has

$$9 \text{ possible total occupancy states of neighboring sites,}$$

$$2 \cdot 9 = 18 \text{ configurations,} \quad 2^{18} = 262,144 \text{ totalistic rules.}$$

This is a large number, but it is exponentially smaller than the number of all possible update rules for the same neighborhood.

### 6.3.1 Conway’s Game of Life

The “game of life” takes its name because it attempts to simulate the reproductive cycle of a species. It is formulated on a square lattice with an update rule involving the eight-cell neighborhood. A new offspring needs exactly three parents in its neighborhood. A living cell dies of loneliness if it has less than two live neighbors, and of overcrowding if it has more than three live neighbors. A living cell feels comfortable with two or three live neighbors; in this case it survives. The complete set of updating rules is listed in Table 6.2.

**Living Isolated Sets** The time evolution of an initial set of a cluster of living cells can show varied types of behavior. Fixpoints of the updating rules, such as a square

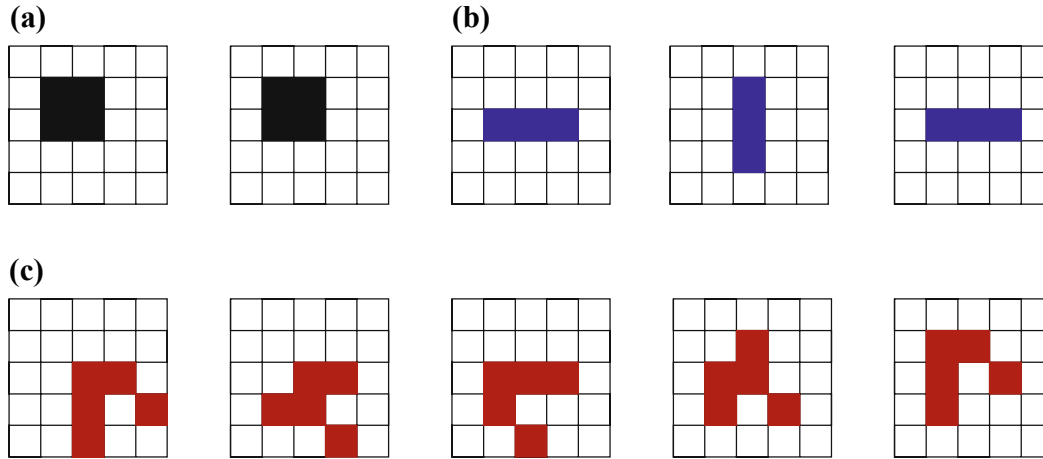
$$\{(0, 0), (1, 0), (0, 1), (1, 1)\}$$

of four neighboring live cells, survive unaltered. There are many configurations of living cells which oscillate, such as three live cells in a row or column,

$$\{(-1, 0), (0, 0), (1, 0)\}, \quad \{(0, -1), (0, 0), (0, 1)\}.$$

**Table 6.2** Updating rules for the game of life, with  $z_i = 0, 1$  corresponding to empty and living cells. An “x” as an entry denotes what is going to happen for the respective number of living neighbors

| $z_i(t)$ | $z_i(t+1)$ | Number of living neighbors |   |   |   |       |
|----------|------------|----------------------------|---|---|---|-------|
|          |            | 0                          | 1 | 2 | 3 | 4...8 |
| 0        | 1          |                            |   |   | x |       |
|          | 0          | x                          | x | x |   | x     |
| 1        | 1          |                            |   | x | x |       |
|          | 0          | x                          | x |   |   | x     |



**Fig. 6.5** Time evolution of some living configurations for the game of life, see Table 6.2. (a) The “block”; it quietly survives. (b) The “blinker”; it oscillates with period 2. (c) The “glider”; it shifts by  $(-1, 1)$  after four time steps

It constitutes a fixpoint of  $f(f(.))$ , alternating between a vertical and a horizontal bar. The configuration

$$\{(0, 0), (0, 1), (0, 2), (1, 2), (2, 1)\}$$

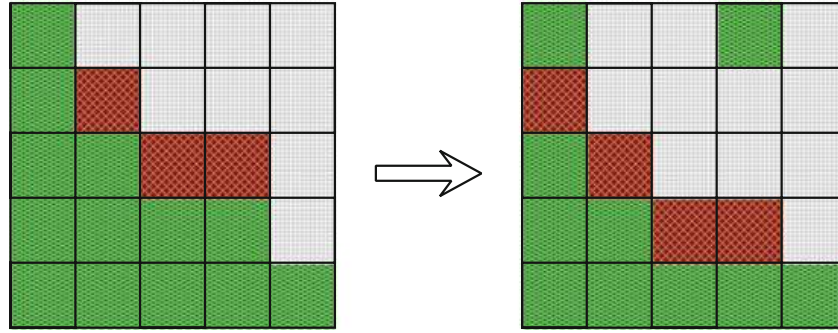
is dubbed “glider”, since it returns to its initial shape after four time steps but is displaced by  $(-1, 1)$ , see Fig. 6.5. It constitutes a fixpoint of  $f(f(f(f(.))))$  times the translation by  $(-1, 1)$ . The glider continues to propagate until it encounters a cluster of other living cells.

**Game of Life as a Universal Computer** It is interesting to investigate, from an engineering point of view, all possible interactions between initially distinct sets of living cells in the game of life. In this context one finds that it is possible to employ gliders for the propagation of information over arbitrary distances. One can prove that arbitrary calculations can be performed by the game of life, when identifying the gliders with bits. Suitable and complicated initial configurations are necessary for this purpose, in addition to dedicated living subconfigurations performing logical computations, in analogy to electronic gates, when hit by one or more gliders.

### 6.3.2 Forest Fire Model

The forest fires automaton is a simplified model of real-world forest fires. It is formulated on a square lattice with three possible states per cell,

$$z_i = 0 \quad (\text{empty}), \quad z_i = 1 \quad (\text{tree}), \quad z_i = 2 \quad (\text{fire}).$$



**Fig. 6.6** For the forest fire model, the time evolution (from left to right) of a configuration of living trees (green), burning trees (red) and of places burnt down (grey). Places burnt down can regrow spontaneously with a small rate, the fire always spreads to nearest neighboring trees

A tree sapling can grow on every empty cell, with probability  $p < 1$ . There is no need for nearby parent trees, as sperms are carried by wind over wide distances. Trees do not die in this model, but they catch fire from any burning nearest neighbor tree. The rules are illustrated in Fig. 6.6.

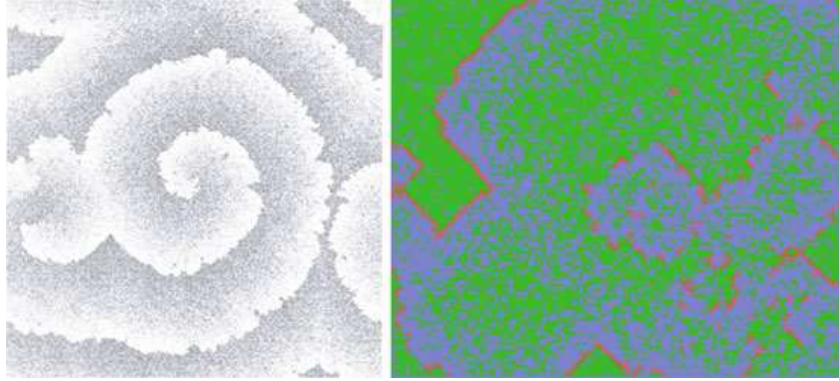
The forest fire automaton differs from typical rules, such as Conway's game of life, because it has a stochastic component. In order to have an interesting dynamics one needs to adjust the growth rate  $p$  as a function of system size, so as to keep the fire burning continuously. The fires burn down the whole forest when trees grow too fast. When the growth rate is too low, on the other hand, the fires, being surrounded by ashes, may die out completely.

| $z_i(t)$ | $z_i(t+1)$ | Condition                  |
|----------|------------|----------------------------|
| Empty    | Tree       | With probability $p < 1$   |
| Tree     | Tree       | No fire close by           |
| Tree     | Fire       | At least one fire close by |
| Fire     | Empty      | Always                     |

When adjusting the growth rate properly, one reaches a steady state, the system having fire fronts continually sweeping through the forest, as it is observed for real-world forest fires, see Fig. 6.7 for an illustration. In large systems stable spiral structures form and set up a steady rotation.

**Criticality and Lightning** As defined above, the forest fire model is not critical, since the characteristic time scale  $1/p$  for the regrowth of trees governs the dynamics. This time scale translates into a characteristic length scale  $1/p$ , which can be observed in Fig. 6.7, via the propagation rule for the fire.

Self-organized criticality can, however, be induced in the forest fire model when introducing an additional rule, namely that a tree might ignite spontaneously with a small probability  $f$ , when struck by lightning, causing also small patches of forest



**Fig. 6.7** Simulations of the forest fire model. **Left:** Fires burn in characteristic spirals for a growth probability  $p = 0.005$  and no lightning,  $f = 0$ . Reprinted from Clar et al. (1996) with permissions, © 1996 IOP Publishing Ltd. **Right:** A snapshot of the forest fire model with a growth probability  $p = 0.06$  and a lightning probability  $f = 0.0001$ . Note the characteristic fire fronts with trees in front and ashes behind

to burn. We will not discuss this mechanism in detail here, treating instead in the next section the occurrence of self-organized criticality in the sandpile model on a firm mathematical basis.

## 6.4 Sandpile Model and Self-Organized Criticality

**Self-Organized Criticality** The notion of ‘life at the edge of chaos’ states,<sup>6</sup> that certain dynamical and organizational aspects of living organisms may be critical. Normal physical and dynamical systems show criticality however only for selected parameters, e.g.  $T = T_c$ , see Sect. 6.1. For criticality to be biologically relevant, the system must evolve into a critical state starting from a wide range of initial states, the defining trait of “self-organized criticality”.

**Sandpile Model** Per Bak and coworkers introduced a simple cellular automaton, the BTW model, that mimics the properties of sandpiles. Every cell is characterized by a force

$$z_i = z(i, j) = 0, 1, 2, \dots, \quad i, j = 1, \dots, L$$

on a finite  $L \times L$  lattice. There is no one-to-one correspondence of the sandpile model to real-world sandpiles. Loosely speaking one may identify the force  $z_i$  with the slope of real-world sandpiles. But this analogy is not rigorous, as the slope

<sup>6</sup> The concept of life at the edge of chaos was first develop in the context of information-processing networks, as discussed Chap. 7.